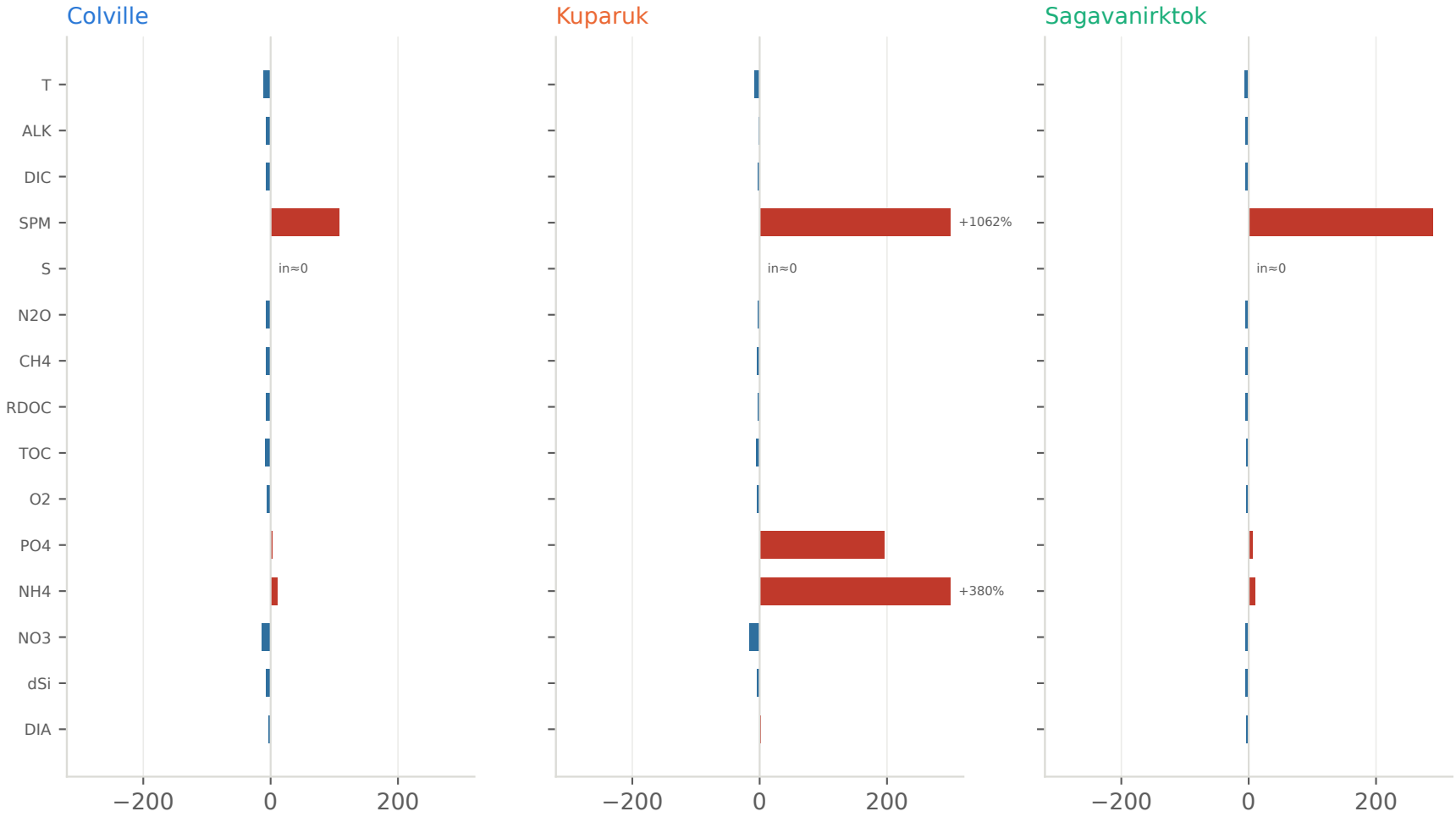
Estuarine filtering — % retained (−) / added (+), 2006–2023 mean



Percent change from input (upstream) to output (mouth) load, mean over 2006–2023, clipped at ±300% for display; 'in≈0' marks a near-zero input base (pct_change is not meaningful there -- see the CSV for the raw value). Orange = net source (added in the estuary), blue = net sink (removed). Upstream (river) boundaries are CONSTANT for every solute except T and TOC -- same limitation as the original 44-yr record. Marine (mouth) boundaries, by contrast, are genuinely time-varying here for EVERY solute (S, DIC, ALK, NO3, NH4, PO4, dSi, O2, TOC -- ECCO-Darwin v5, 2005-2023), which is new relative to the original figures (only T was marine-time-varying there). Structure seen here reflects BOTH the model's own internal dynamics acting on a constant upstream source AND a genuinely time-varying ocean signal propagating in from the mouth. See CLAUDE.md -> "Interannual forcing" -> "Full-forcing interannual (2005-2023)". This is an ADVECTIVE-flux-only estimate. For S (salinity) specifically, dispersive/tidal-pumping transport -- the dominant salt-intrusion mechanism on this coast (see CLAUDE.md -> "Width and dispersion") -- is NOT captured, so S's entry likely understates true salt exchange; the biogeochemical solutes are advection-dominated and less affected by this limitation.